

Weight gain over 5 years in 21,966 meat-eating, fish-eating, vegetarian, and vegan men and women in EPIC-Oxford.



BACKGROUND: Cross-sectional studies have shown that vegetarians and vegans are leaner than omnivores. Longitudinal data on weight gain in these groups are sparse. OBJECTIVE: We investigated changes in weight and body mass index (BMI) over a 5-year period in meat-eating, fish-eating, vegetarian, and vegan men and women in the UK. DESIGN: Self-reported anthropometric, dietary and lifestyle data were collected at baseline in 1994-1999 and at follow-up in 2000-2003; the median duration of follow-up was 5.3 years. SUBJECTS: A total of 21,966 men and women participating in Oxford arm of the European Prospective Investigation into Cancer and Nutrition aged 20-69 years at baseline. RESULTS: The mean annual weight gain was 389 (SD 884) g in men and 398 (SD 892) g in women. The differences between meat-eaters, fish-eaters, vegetarians and vegans in age-adjusted mean BMI at follow-up were similar to those seen at baseline. Multivariable-adjusted mean weight gain was somewhat smaller in vegans (284 g in men and 303 g in women, $P < 0.05$ for both sexes) and fish-eaters (338 g, women only, $P < 0.001$) compared with meat-eaters. Men and women who changed their diet in one or several steps in the direction meat-eater --> fish-eater --> vegetarian --> vegan showed the smallest mean annual weight gain of 242 (95% CI 133-351) and 301 (95% CI 238-365) g, respectively. CONCLUSION: During 5 years follow-up, the mean annual weight gain in a health-conscious cohort in the UK was approximately 400 g. Small differences in weight gain were observed between meat-eaters, fish-eaters, vegetarians and vegans. Lowest weight gain was seen among those who, during follow-up, had changed to a diet containing fewer animal food.